

Chapter 4 Overview

4.1 System of Differential Equations	
Learning Outcomes	- What is a first order system of differential equations (SDE)? - What is an initial value problem? - How to convert a second order ODE to a SDE? - How to set up a model of 2 x 2 SDE for mixing problem? - How to set up a model for mass-spring problem?
Key Points	- A solution of a single differential equation is a function. A solution of a system of differential equations is a vector function. - A (linear) SDE can be expressed in matrix form $\mathbf{y}'(t) = \mathbf{A} \mathbf{y}(t)$ - A second order ODE can be converted into a first order SDE - The motion of an undamped mass-spring system is modeled by $y'' + c y = 0$ - The motion of a (damped) mass-spring system is modeled by $y'' + ay' + by = 0$
4.2. Solutions of System of Differential Equations	
Learning Outcomes	- How to find solutions of SDE $\mathbf{y}'(t) = \mathbf{A} \mathbf{y}(t)$? - How are eigenvalues and eigenvectors of $\mathbf{A}$ related to the solutions of $\mathbf{y}'(t) = \mathbf{A} \mathbf{y}(t)$? - What is superposition principle? - What is a fundamental set of solutions of an SDE?
Key Points	- If r and $\mathbf{v}$ are corresponding eigenvalue and eigenvector of $\mathbf{A}$, then $\mathbf{y}(t) = \mathbf{v} e^{rt}$ is a solution of the SDE $\mathbf{y}'(t) = \mathbf{A} \mathbf{y}(t)$. - If $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$ are solutions of an SDE, then any linear combination $c_1 \mathbf{x}_1(t) + c_2 \mathbf{x}_2(t)$ is also a solution. - The general solution of an SDE is given by the linear combinations of a fundamental set of solutions. - If the matrix $\mathbf{A}$ for the SDE $\mathbf{y}'(t) = \mathbf{A} \mathbf{y}(t)$ has two linearly independent eigenvectors, then it has a fundamental set of solutions. - There is exactly one solution that satisfies a given SDE with a given initial condition.
4.3. Phase Plane	
Learning Outcomes	- What are the graphical representation of a vector function? - What are the phase plane and trajectories of an SDE? - How to sketch trajectories of an SDE? - What is an equilibrium point of an SDE? - How to determine the direction of the trajectories?
Key Points	- Each particular solution of an SDE correspond to one trajectory on the phase plane. - The tangent $\mathbf{y}'$ at each point $\mathbf{y}$ of the trajectory can be computed easily using $\mathbf{y}' = \mathbf{A} \mathbf{y}$ $\mathbf{y} = \mathbf{0}$ is a solution of any SDE corresponding to an equilibrium point of the SDE. - The behavior of trajectories is determined by the powers of the exponential functions in the solutions.

4.4. Classification of Equilibrium Points	
Learning Outcomes	• What are the different types of equilibrium points of 2×2 SDE? • How do the eigenvalues and eigenvectors of the underlying matrix of an SDE determine the types of the equilibrium points? • How does the phase portrait look like for each corresponding type of equilibrium point? • What are the different types of stability of equilibrium points? • What does it mean by an equilibrium point is stable/unstable? • How do the eigenvalues determine the stability of an equilibrium point? • How does the stability of the equilibrium point of the SDE for the mass-spring system correspond to the damping condition?
Key Points	• There are six types of equilibrium points: node, star node, degenerate node, saddle point, center and spiral point. • The types of equilibrium points are completely determined by the eigenvalues of the underlying matrix of the SDE • If the underlying matrix has 0 as eigenvalue, then it has infinitely many equilibrium points along the line parallel to the eigenvector of this 0 eigenvalue. • If 0 is not an eigenvalue of the underlying matrix, then it has only one equilibrium point at the origin. • There are three types of stability: asymptotically stable, neutrally stable, and unstable. • Center is the only equilibrium point that is neutrally stable. • Saddle points are always unstable. • Node and spiral points can be unstable or asymptotically stable. • An equilibrium point is asymptotically stable only when both eigenvalues are negative. • An equilibrium point is unstable if at least one eigenvalue is positive.
4.5. Real Distinct Eigenvalues	
Learning Outcomes	• What type of equilibrium point can it be when the underlying matrix of the SDE has real distinct eigenvalues? • How do the trajectories behave when the two distinct eigenvalues have opposite signs? • How do the trajectories behave when the two distinct eigenvalues have same sign?
Key Points	• When the eigenvalues have opposite signs, the equilibrium point is a saddle point. • When the eigenvalues have same sign, the equilibrium point is a node.

	4.6. Complex Eigenvalues
Learning Outcomes	• How to find eigenvectors associated with complex eigenvalues? • How to get a fundamental set of real solutions for a system with complex eigenvalues? • What type of equilibrium point can it be when the underlying matrix of the SDE has complex eigenvalues? • How do the trajectories behave when the complex eigenvalues are purely imaginary? • How do the trajectories behave when the complex eigenvalues have non-zero real parts?
Key Points	• For matrices with complex eigenvalues, the eigenvalues and the corresponding eigenvectors always come in conjugate pairs. • When the eigenvalues are purely imaginary, the equilibrium point is a center. • When the eigenvalues have non-zero real parts, the equilibrium point is a spiral point. • The two real solutions $\mathbf{x}_{\text{Re}}(t)$ and $\mathbf{x}_{\text{Im}}(t)$ derived from the real and imaginary parts of the eigenvalue and eigenvectors are linearly independent and gives a fundamental set of solutions for the system.
	4.7. Repeated Eigenvalues
Learning Outcomes	• What type of equilibrium point can it be when the underlying matrix of the SDE has a repeated eigenvalue? • How to find a second solution of an SDE when the underlying matrix has only one independent eigenvector? • What is a generalized eigenvector and how to find it? • How do the trajectories behave when the repeated eigenvalue is associated with two linearly independent eigenvectors? • How do the trajectories behave when the repeated eigenvalue is associated with only one linearly independent eigenvector?
Key Points	• If a matrix $\mathbf{A}$ has only one eigenvalue r and correspond to only one eigenvector $\mathbf{v}$, a generalized eigenvector $\mathbf{u}$ can be found by solving the non-homogeneous system $(\mathbf{A} - r\mathbf{I})\mathbf{u} = \mathbf{v}$. • The generalized eigenvector can be used to construct a second independent solution for the SDE. • When the repeated eigenvalue is associated with two linearly independent eigenvectors, the equilibrium point is a star node. • When the repeated eigenvalue is associated with only one linearly independent eigenvector, the equilibrium point is a degenerated node. • Trajectories on opposite side of the eigenvectors on the phase plane move in opposite directions.